Mehmet A. Guler

Contact Information

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Academic Profile

Professor of Mechanical Engineering with extensive undergraduate and graduate teaching, design-project mentoring, professional engineering instruction, and industrial structural and thermal analysis experience. Teaching spans mechanics, materials, Machine Design I and II, Engineering Design, System Modeling and Analysis, finite element methods, and thermal-fluid fundamentals. Supervised four OYAK Renault-supported senior design projects and delivered ANSYS instruction for university and professional audiences. Experience in curriculum development, course coordination, ABET-related assessment, and faculty development supports collaborative contributions to teaching, student success, and academic service. Research in solid and computational mechanics provides applied examples for instruction and student projects.

Education

Lehigh University, **Bethlehem, PA, US**
Ph.D., Mechanical Engineering
Dissertation: *Contact Mechanics of FGM Coatings*
Advisor: Fazil Erdogan

Lehigh University, **Bethlehem, PA, US**
M.S., Mechanical Engineering

University of Sussex, **Brighton, UK**
M.S., Computer Technology in Manufacturing

Middle East Technical University, **Ankara, Turkey**
B.S., Mechanical Engineering

Academic Experience

American University of the Middle East (AUM), **Kuwait**
Professor **Sep. 2018 – Present**
Served as Head of the Research and Faculty Development Committee, 2018–2023.
Taught undergraduate courses across core mechanical engineering, including Statics, Mechanics of Materials, Engineering Materials, Machine Design, Engineering Design, System Modeling and Analysis, Thermodynamics, Fluid Mechanics, and Hydraulics.

TOBB University of Economics and Technology, **Ankara, Turkey**
Professor **Oct. 2015 – Sep. 2018**
Associate Professor **Jul. 2010 – Oct. 2015**
Assistant Professor **Sep. 2006 – Jul. 2010**
Taught courses from sophomore through graduate level in statics, dynamics, machine design, finite element methods, engineering mathematics, and theory of elasticity.

Near East University, **Nicosia, N. Cyprus**
Professor (Part-time) **Sep. 2014 – Sep. 2018**
Taught junior-level machine design courses.

University of Arizona, **Tucson, AZ, US**
Visiting Research Scholar **Feb. 2014 – Jun. 2014**
Conducted research on modeling hyperelastic materials using peridynamic theory for a project funded by the Department of Aerospace and Mechanical Engineering at the University of Arizona.
Fulbright Research Scholar **Aug. 2013 – Feb. 2014**
Conducted research on modeling composite materials using peridynamic theory for the project titled "Development of a method to predict strength and failure of layered composites using peridynamic theory," funded by Fulbright.

Virginia Commonwealth University, **Richmond, VA, US**
Visiting Research Scholar **Jun. 2007 – Sep. 2007**
Conducted research on springback prediction of Advanced High Strength Steels used in automobile components.

Lehigh University, **Bethlehem, PA, US**
Research Assistant **Jan. 1994 – Sep. 2000**
Research focused on stress analysis and fracture characterization of ceramic coatings on metal substrates, involving mechanical and thermal stresses. Developed analytical models to study the contact mechanics of functionally graded materials.

Teaching Experience and Interests

Design and capstone instruction: Machine Design I and II, Mechanical System Design, Engineering Design, and undergraduate project supervision. Industry-sponsored senior-design projects include Renault support-bracket dynamics, bumper energy absorption, cockpit-carrier crash analysis, and side-crash simulation; project details and funding appear under Funded Research and Industry-Sponsored Design Projects.

Core mechanical engineering instruction: Statics, Dynamics, Mechanics of Materials, Engineering Materials, System Modeling and Analysis, Thermodynamics, Fluid Mechanics, Hydraulics, and Modern Mechanics. Graduate instruction includes Finite Element Methods, Engineering Mathematics, and Theory of Elasticity.

Simulation and professional education: ANSYS Mechanical APDL and Workbench instruction connects mathematical models with physical response, verification, and design decisions. Industrial experience provides examples of structural, thermal, dynamic, contact, and fatigue analysis.

Instructional interests: Undergraduate and graduate mechanics, materials, mechanical design, and engineering analysis; project-based learning linking calculations with manufacturability, testing, and technical communication. Interested in contributing to applied manufacturing instruction and advanced computational analysis electives in collaboration with departmental colleagues.

Large-class teaching and student engagement: Use ClassPoint for conceptual questions and student responses. Planned extensions include Think–Pair–Share, peer instruction, and low-stakes retrieval quizzes, with clear rubrics and individual explanations within team projects to assess understanding. These proposed activities would support participation and targeted feedback across different levels of preparation.

Student Mentoring

Undergraduate Design and Project Mentoring
Advised senior engineering design projects with industrial and computational components. Mentoring addresses problem formulation, engineering analysis, design development, verification, interpretation, and technical communication. Renault-funded senior-design projects provided experience connecting student investigations with industrial engineering questions. Teaching experience includes Engineering Design and repeated offerings of Machine Design I and II.

Graduate Research Supervision
Supervised and co-supervised Ph.D. and M.S. students in computational mechanics, structural analysis, crashworthiness, fatigue, composites, peridynamics, finite element analysis, and mechanical design. Detailed supervision records appear later in this CV.

Courses Taught

AUM: American University of the Middle East
TOBB ETU: TOBB University of Economics and Technology
NEU: Near East University
Year ranges indicate multiple offerings, not continuous teaching.

Mechanical Design and Engineering Applications

Course	Institution / Code	Years
Machine Design I	AUM / ME 352	2018–2026
	NEU / ME 303	2014–2016
Machine Design II	AUM / ME 452	2018–2026
	NEU / ME 304	2015–2017
Machine Elements	TOBB ETU / MAK 312	2008–2015
Mechanical System Design	TOBB ETU / MAK 404	2009–2012
Engineering Design	AUM / ME 463	2020–2026
Automotive Engineering	TOBB ETU / MAK 444	2013, 2016

Mechanics and Materials

Statics	TOBB ETU / MAK 104	2007–2017
	AUM / CVL 297	2024–2025
Basic Mechanics I	AUM / ME 270	2020–2023
Dynamics	TOBB ETU / MAK 203	2006
Strength of Materials	TOBB ETU / MAK 206	2006–2017
Mechanics of Materials	AUM / ME 323	2018–2020
Engineering Materials I	AUM / CE 231	2026
Modern Mechanics	AUM / PHYS 172	2025–2026

Computational Methods and System Modeling

Finite Element Methods	TOBB ETU / MAK 410	2008
System Modeling and Analysis	AUM / ME 375	2024–2026

Thermal and Fluid Sciences

Thermodynamics I	AUM / ME 200	2025
Fluid Mechanics	AUM / ME 309	2024
Hydraulics	AUM / CVL 340	2025

Graduate Instruction

Engineering Mathematics	TOBB ETU / MAK 501	2010–2016
Theory of Elasticity	TOBB ETU / MAK 506	2009–2010
Finite Element Methods	TOBB ETU / MAK 510	2007, 2016

Professional Training and Short Courses Taught

University of Modena and Reggio Emilia, **Modena, Italy**
Instructor – Contact Mechanics with ANSYS Mechanical APDL **Jul. 2025 – Sep. 2025**
Delivered a 36-hour specialized course on contact mechanics using ANSYS Mechanical APDL.

UCTEA Chamber of Mechanical Engineers, **Ankara, Turkey**
ANSYS Workbench Instructor **Mar. 2010 – Feb. 2013**
Delivered professional training in static, dynamic, harmonic, modal, fatigue, and thermal finite element analysis using ANSYS Workbench.

Academic Leadership and Service

American University of the Middle East (AUM) Kuwait

Head, Design Focus Group 2025–Present
Lead and coordinate curriculum assessment and continuous improvement activities related to design courses, Student Outcomes, and accreditation requirements.

Head, Research and Faculty Development Committee 2018–2023
Led research planning and faculty-development activities, including seminars, research-project monitoring, and support of scholarly activities.

Member, Accreditation Committee 2024–Present
Contribute to accreditation documentation, assessment of Student Outcomes and Program Educational Objectives, and continuous program improvement.

Member, Curriculum Committee 2024–Present
Contribute to curriculum review, course and program development, and accreditation-related continuous improvement.

Head, Learning and Development Committee Fall 2026–Present
Support learner-centered activities, capstone design projects, program learning goals, and extracurricular academic activities.

TOBB University of Economics and Technology Ankara, Turkey

Head, Mechanics Focus Group 2006–2018
Coordinated curriculum, teaching, and academic activities in mechanics and related areas.

Professional Experience

SANKO / MST Construction Equipment, **Gaziantep, Turkey**
Engineering Consultant **2016 – 2018**

- Provided structural analysis and fatigue engineering consultancy for welded telescopic booms used in MST telehandlers. Supported finite element analysis and fatigue-life assessment, evaluation of critical welded regions using Eurocode 3, BS 7608, IIW, and FKM methodologies, and correlation of numerical predictions with strain-gauge measurements and static/fatigue testing.

The Scientific and Technological Council of Turkey (TÜBİTAK), **Ankara, Turkey**
Executive Committee Member for MAKİTEG Group **Apr. 2017 – Sep. 2018**

- Served as a committee member evaluating research projects submitted by manufacturing industry to the Machinery and Manufacturing Technologies Group (MAKİTEG).

Executive Committee Member for USETEG Group **Apr. 2015 – Apr. 2016**

- Served as a committee member evaluating research projects submitted by automotive industry to the Transportation, Defense, Energy and Textile Technologies Group (USETEG).

Arçelik Dishwasher Inc., **Ankara, Turkey**
Project Consultant **Sep. 2013 – Sep. 2015**

- Principal investigator of the industrial project titled "Optimization of the structural parts and packaging of a dishwasher using Finite Element Method," funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK TEYDEB Project number 5150016).

Oyak Renault Inc., **Bursa, Turkey**
Project Consultant **Sep. 2011 – Jun. 2013**

- Supported Renault engineers in finite element side-crash analysis for the project titled "Renault Fluence mid-floor part monolithic design and production of the prototype by hot forming technology," funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK TEYDEB Project number 3110357).
- Performed dynamic analysis of a VSP support bracket.
- Designed a shock-absorbing component for a front bumper.
- Conducted frontal crash analysis for the design of an automobile cockpit carrier.

GE Marmara Technology Center (GE MTC) Inc., **Gebze, Turkey**
Consultant **Jan. 2010 – Jan. 2011**

- Supported GE MTC engineers in finite element analysis of aircraft-engine components using ANSYS.

TEMSA GLOBAL Inc., **Gebze, Turkey**
Project Consultant **Sep. 2008 – Sep. 2010**

- Principal investigator of the industrial project titled "Development of passive safety system for absorbing crash energy during frontal crashes involving intercity busses," funded by the Ministry of Industry and Trade of Turkey under the SAN-TEZ program.

TEMSA GLOBAL Inc., **Adana, Turkey**
Senior Structural Engineer **Mar. 2005 – Sep. 2006**

- Managed all phases of bus rollover simulation projects (ECE R66), including building finite element models in ANSA, preparing mass lists, verifying analysis data such as centers of gravity and mass moments of inertia of rigid components, setting up LS-DYNA analyses, running simulations, verification, and certification.
- Performed three-dimensional stress analysis of automotive parts under structural and dynamic loads using ANSYS, identified critical locations, and recommended design changes to improve component life.
- Carried out modal analysis of buses and created animations illustrating modes of excitation.

CD-ADAPCO Inc. (Now SIEMENS), **Melville, NY, US**
Senior Structural and Thermal Engineer **Sep. 2000 – Sep. 2004**

- Reviewed customer contracts to determine analysis requirements, goals, and appropriate methods.
- Modeled a glass-lined reactor and performed structural stress analysis to investigate possible causes of glass-lining failure.
- Conducted single-cylinder thermal analysis of a six-cylinder diesel engine to understand the thermal behavior of the cylinder and head.
- Performed three-dimensional stress analysis of rotating parts in gas turbines and compressors under thermal, structural, and dynamic loads using ANSYS.
- Carried out modal analysis of a generator and created an animation illustrating modes of excitation.
- Performed thermal and structural analysis of EGR coolers used in truck engines, identified regions susceptible to cracking, and recommended design changes to improve component life.
- Developed models involving thermal and structural loading, intermittent contact, friction, and nonlinear material properties.
- Created programs and scripts to automate model generation, analysis setup, and verification.

Research Areas

Contact and interface mechanics; functionally graded materials and coatings; size-dependent elasticity; structural stability and thin-walled structures; crashworthiness and energy absorption; fracture, fatigue, and composite damage; finite element and peridynamic modeling; manufacturing simulation and springback; data-assisted modeling and mechanical design optimization.

Funded Research and Industry-Sponsored Design Projects

E. Acar and M.A. Guler (2019–2021), "Design and optimization of a new energy efficient cylindrical roller bearing using contact mechanics and optimization techniques," funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $35,000.

M.A. Guler and D. Coker (2017–2020), "Numerical and experimental investigation of fracture propagation in composites with impact damage," funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $100,000.

B. Yıldırım and M.A. Guler (2017–2018), "Design and optimization of the body and chassis of a new tram vehicle under dynamic and crash loads using Finite Element Method," funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $50,000.

M.A. Guler (2015–2016), "Examination of impact energy absorbing capacity of empty and aluminum based foam-filled crash boxes," funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $10,000.

M.A. Guler (2013–2015), "Optimization of the structural parts and packaging of a dishwasher using Finite Element Method," funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $135,000.

M.A. Guler and B. Aksoylu (2013–2015), "Development of a method to predict strength and failure of layered composites using peridynamic theory." Budget $125,000.

M.A. Guler (2013), "Dynamic Analysis of a VSP Support Bracket as a senior design project," funded by OYAK RENAULT. Budget $1,500.

M.A. Guler (2013), "Design of a Shock Absorbing Component for a Front Bumper of an Automobile as a senior design project," funded by OYAK RENAULT. Budget $1,500.

M.A. Guler (2012), "Frontal crash analysis for the design of an automobile cockpit carrier as a senior design project," funded by OYAK RENAULT. Budget $1,500.

M.A. Guler, R.M. Görgülüarslan and Fırat Özer (2011), "Side crash simulation analysis for monolithic production of reinforced parts as a senior design project," funded by OYAK RENAULT. Budget $3,000.

M.A. Guler and E. Acar (2009–2011), "Accurate prediction and robust optimization of springback in dual phase steels during sheet metal forming operations," funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $65,000.

M.A. Guler (2008–2010), "Development of passive safety system for absorbing crash energy during frontal crashes involving intercity busses," funded by the Ministry of Industry and Trade of Turkey under the SAN-TEZ program. Budget $225,000.

M.A. Guler and S. Dag (2007–2009), "Computational and Analytical methods for contact mechanics analysis of functionally graded material," funded by The Scientific and Technical Research Council of Turkey (TÜBİTAK). Budget $70,000.

Scholarly Recognition and Editorial Appointments

Listed in Top 2% Scientists Worldwide by Stanford University, since 2020.

Editorial Board Member, Thin-Walled Structures, ELSEVIER, since 2018.

Associate Editor, Measurement and Control, SAGE, since 2024.

Editorial Review Board, Journal of Applied Biomaterials & Functional Materials, SAGE, since 2025.

Fulbright Visiting Research Scholarship to University of Arizona, 2013.

The most cited articles for the paper entitled "The effect of geometrical parameters on the energy absorption characteristics of thin-walled structures under axial impact loading" in International Journal of Crashworthiness.

The most cited articles published for the paper entitled "Multi-objective crashworthiness optimization of tapered thin-walled tubes with axisymmetric indentations" in Thin-Walled Structures.

The most cited articles published from 2004 to 2008 for the paper entitled "Contact mechanics of graded coatings" in International Journal of Solids and Structures.

Invited Talks and Keynote Presentations

Keynote Speaker — 3rd IC-EMAMM 2026 **Sep. 12, 2026**
Computational Mechanics and Finite Element Analysis for the Design and Optimization of Lightweight Structures
3rd International Conference on Energy, Manufacture, Advanced Material, and Mechatronics, Makassar, Indonesia.

Postdoctoral Researcher

A. Kucuksucu, "Contact mechanics of orthotropically graded materials," 2012–2014.

Visiting Research Scholar

Y. Alinia, "Rolling contact mechanics of graded materials," 2011–2012.

Supervised Ph.D. Students

S.K. Mert (2022), "Experimental and Numerical Investigation of Energy Absorbing Capacity of Aluminum, Composite, and Hybrid Crash Boxes." (Co-supervised with Dr. Adem Çiçek)

O. Mulkoglu (2022), "Fatigue Life Estimation of Welded Structures: A New Modified Hot-spot Stress Method." (Co-supervised with Dr. Fatih Göncü)

U. Yolum (2021), "Investigation of bending and mode II failures in thick plates using peridynamic theory."

E. Ince (2019), "Design of a new power-split infinitely variable transmission."

M. Altın (2017), "Investigation of the energy absorption capacities of metallic foam filled crash-boxes used in automobiles." (Co-supervised with Dr. H.S. Yücesu)

Supervised M.Sc. Students

S. Eltaş (2023), "Finite Element Analysis of Beam-to-Beam Eccentric End Plate Connections." (Co-supervised with Dr. Bora Yıldırım)

S.A. Keskin (2021), "Studying and optimizing the effects of different design options on the performance of crash absorbers with diagrams." (Co-supervised with Dr. Erdem Acar)

B.B. Mercan (2021), "Investigation of holed composite plates exposed to in-plane loading by the peridynamic theory." (Co-supervised with Dr. Bora Yıldırım)

E. Gök (2020), "Investigation of damage behavior in composite materials under mode I and mode II loads using peridynamic theory."

M. Bektaş (2020), "Aeroelastic analysis of bio-inspired flapping wing models." (Co-supervised with Dr. D.F. Kurtuluş)

B. Yılmaz (2019), "Application of multi-fidelity modeling technique on crash analyses." (Co-supervised with Dr. E. Acar)

B.M. Zeytinci (2019), "Investigation of the effect of the ring type stiffeners on the Vierendeel mechanism of perforated beams under bending."

E. Özaslan (2019), "Investigation of mechanical strength of open hole composite plates with finite element method."

A.R. Seyedi (2019), "Design and optimization for body and chassis of new tram with finite element method under static, dynamic and collision loads." (Co-supervised with Dr. B. Yıldırım)

F. Can (2018), "The design of Flow controller unit for Joule-Thomson type cryocooler." (Co-supervised with Dr. M.K. Aktaş)

A.E. Örün (2017), "Effect of hole reinforcement on buckling behavior under combined loads for a beam structure."

K.B. Yılmaz (2016), "Finite element analysis of contact mechanics of rigid punches sliding over a graded orthotropic medium." (Co-supervised with Dr. B. Yıldırım)

N. Tüten (2016), "Weight optimization and reliability prediction of an automobile torque arm subjected to cyclic loading." (Co-supervised with Dr. E. Acar)

A. Tastan (2016), "Development of a method to predict strength and failure of isotropic and composite structures under tension and bending loadings using bond-based peridynamic theory."

O. Mülkoğlu (2016), "Optimization of the mechanical structure of a dishwasher and its packaging module using Finite Element Method."

T. Kiper (2015), "Analysis of bird strike effects on the wing of a training aircraft." (Co-supervised with Dr. İ. Uslan)

O.M. Bircan (2014), "An analytical approach to design form-turning insert profiles: Investigation of rake face design on tool life." (Co-supervised with Dr. Y. Karpat)

A.T. Camcı (2013), "Improvements on the range and the performance of a light electric vehicle by making use of a continuously variable transmission."

F. Özer (2011), "Springback prediction in Advanced High Strength Steels after forming operations." (Co-supervised with Dr. İ.B. Özsoy)

D. Bekar (2011), "Robust optimization of springback predictions in dual phase steels during sheet metal forming operations." (Co-supervised with Dr. E. Acar)

L. Sözen (2011), "Springback prediction in rotary tube bending operations."

M.E. Cerit (2011), "Development of passive safety system for absorbing crash energy during frontal crashes involving intercity busses."

Y.F. Gülver (2009), "Mechanical modeling of thin films bonded to functionally graded coatings."

External Thesis Examination Committees

Cong Tien Nguyen, Naval Architecture, Ocean and Marine Engineering, University of Strathclyde,
Ph.D. thesis defence external committee member, November 2020.
Title of the thesis: Peridynamics for damage prediction in ships and offshore structures.

Tarek Salaheldin Shaban, Department of Mechanical Engineering, Qatar University,
M.Sc. thesis defence external committee member, April 2019.
Title of the thesis: Stress Shielding Reduction Approach for Femoral Stems used in Total Hip Arthroplasty (THA) using topology optimization.

Al-Sendibad Said Shehadeh, Department of Mechanical Engineering, Qatar University,
M.Sc. thesis defence external committee member, April 2019.
Title of the thesis: Evaluation of 316L Stainless Steel Part fabrication using additive and subtractive manufacturing: a guideline for process selection.

Onur Arslan, Department of Mechanical Engineering, Middle East Technical University,

Ph.D. thesis defence external committee member, June 2017.

Jabbar Gattmah, Department of Mechanical Engineering, Yıldırım Beyazıt University,
Ph.D. thesis defence external committee member, June 2017.

Publications

Journal Papers

E Radi and MA Güler. Pin–hole contact with curvature-induced couple tractions in couple-stress elasticity. *International Journal of Solids and Structures*, page 114121, 2026

E Radi and MA Güler. Effects of microstructure in pin-loaded hole contact with clearance. *International Journal of Engineering Science*, 221:104454, 2026

Özlem Akçay, Murat Altın, Mehmet Ali Güler, and Erdem Acar. Machine Learning Based Crashworthiness Optimization of 3D-Printed Hybrid Multi-Cell Energy Absorber. *European Journal of Mechanics-A/Solids*, page 106109, 2026

Fei Shen, You-Hua Li, Mehmet Ali Güler, Hong-Dong Wu, Wei-Wei Shen, and Liao-Liang Ke. A high-efficiency prediction method for thermal contact resistance of rough surfaces. *International Communications in Heat and Mass Transfer*, 167:109325, 2025

You-Hua Li, Liao-Liang Ke, Gang-Gang Chang, Mehmet Ali Güler, and Fei Shen. On the electrical contact resistance model of rough surfaces based on multi-physics modeling. *International Journal of Solids and Structures*, page 113583, 2025

You-Hua Li, Liao-Liang Ke, Kun Zhou, Gang-Gang Chang, Mehmet Ali Güler, Wei-Wei Shen, and Fei Shen. On the multi-physics elastoplastic electrical contact of rough surfaces. *Tribology International*, 204:110418, 2025

Ali Mamedov, Ali Dinc, Mehmet Ali Guler, Murat Demiral, and Murat Otkur. Tool wear in micromilling: a review. *The International Journal of Advanced Manufacturing Technology*, 137(1):47–65, 2025

Lovejoy Mutswatiwa, Mehmet Ali Güler, and Celestin Nkundineza. An Investigation on the Impact of Non-Uniform Track Stiffness on Wheel-Rail Interaction Mechanics. *Journal of Vibration Engineering & Technologies*, 13(7):518, 2025

Ahmed Saber, Mehmet Ali Güler, Erdem Acar, Omar Soliman ElSayed, Hussain Aldallal, Abdulrahman Alsadi, and Yousef Aldousari. Crash performance of additively manufactured tapered tube crash boxes: influence of material and geometric parameters. *Designs*, 9(3):72, 2025

Oguzhan Mulkoglu, Mehmet Ali Güler, and Fatih Göncü. Modified hot-spot stress method for fatigue life estimation of welded components. *Materials Testing*, 67(5):821–832, 2025

Mehmet Ali Güler and Yadolah Alinia. An elastic-constant stress cohesive zone model for an orthotropic piezoelectric actuator adhesively bonded to a substrate. *Engineering Fracture Mechanics*, page 110391, 2024

Ahmed Saber, Mehmet Ali Güler, Murat Altin, and Erdem Acar. Bio-inspired thin-walled energy absorber adapted from the xylem structure for enhanced vehicle safety. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 46(10):613, 2024

You-Hua Li, Fei Shen, Mehmet Ali Güler, and Liao-Liang Ke. An efficient method for electro-thermo-mechanical coupling effect in electrical contact on rough surfaces. *International Journal of Heat and Mass Transfer*, 226:125492, 2024

Dengke Li, Peijian Chen, Hao Liu, Zhilong Peng, Mehmet Ali Guler, and Shaohua Chen. The interfacial behavior of an axisymmetric film bonded to a graded inhomogeneous substrate. *Mechanics of Materials*, page 104983, 2024

Erdem Acar, Evren Emre Üstün, Mehmet Ali Güler, Ahmet Toros, and Ozan Müştak. Design of an energy efficient cylindrical roller bearing. *Mechanics Based Design of Structures and Machines*, 52(2):826–847, 2024

You-Hua Li, Fei Shen, Mehmet Ali Güler, and Liao-Liang Ke. Modeling multi-physics electrical contact on rough surfaces considering elastic-plastic deformation. *International Journal of Mechanical Sciences*, page 109066, 2024

Mehmet Ali Güler, Yadolah Alinia, and Enrico Radi. Couple Stress Effects in a Thin Film Bonded to a Half-space. *Mathematics and Mechanics of Solids*, 29(4):629–644, 2024

You-Hua Li, Fei Shen, Mehmet Ali Güler, and Liao-Liang Ke. A rough surface electrical contact model considering the interaction between asperities. *Tribology International*, 190:109044, 2023

Mehmet Ali Güler, Sinem K Mert, Murat Altin, and Erdem Acar. An investigation on the energy absorption capability of aluminum foam-filled multi-cell tubes. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 45(10):541, 2023

Mehmet Ali Güler, Yadolah Alinia, and Ergun Nart. Interfacial delamination for an orthotropic thin film/substrate system. *Theoretical and Applied Fracture Mechanics*, page 104007, 2023

Seyed Ali Abbaszadeh-Fathabadi, Yadolah Alinia, and Mehmet Ali Güler. On the mechanics of a double thin film on a finite thickness substrate. *International Journal of Solids and Structures*, page 112349, 2023

E Radi, A Nobili, and MA Guler. Indentation of a free beam resting on an elastic substrate with an internal lengthscale. *European Journal of Mechanics-A/Solids*, page 104804, 2022

Sinem K Mert, Mehmet Ali Güler, Murat Altin, Erdem Acar, and Adem Çiçek. Crashworthiness evaluation of hybrid tubes made of GFRP composite and aluminum tubes filled with honeycomb structures. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 45(6):291, 2023

Samet Eltaş, Mehmet Ali Güler, Konstantinos Daniel Tsavdaridis, Christos E Sofias, and Bora Yıldırım. On the beam-to-beam eccentric end plate connections: A numerical study. *Thin-Walled Structures*, 188:110787, 2023

Qing-Hui Luo, Yue-Ting Zhou, and Mehmet Ali Guler. Adhesive behavior between dissimilar materials subjected to thermo-elastic loadings with normal-tangential coupling effect. *Applied Mathematical Modelling*, 115:360–384, 2023

Ugur Yolum, Mirac Onur Bozkurt, Eda Gok, Demirkan Coker, and Mehmet Ali Güler. Crack propagation in the Double Cantilever Beam using Peridynamic theory. *Composite Structures*, page 116050, 2022

Oguzhan Mulkoglu, Alireza Seyedi, Mehmet Ali Güler, Bora Yildirim, and Fatih Göncü. Fatigue life estimation of welded connections in a tram vehicle. *International Journal of Heavy Vehicle Systems*, 29(5):537–552, 2022

Cüneyt Aktaş, Erdem Acar, Mehmet Ali Güler, and Murat Altın. An investigation of the crashworthiness performance and optimization of tetra-chiral and reentrant crash boxes. *Mechanics Based Design of Structures and Machines*, pages 1–24, 2022

Yadolah Alinia and Mehmet Ali Güler. On the problem of an axisymmetric thin film bonded to a transversely isotropic substrate. *International Journal of Solids and Structures*, 248:111636, 2022

İsa Çömez and Mehmet Ali Güler. Thermoelastic contact problem of a rigid punch sliding on a functionally graded piezoelectric layered half plane with heat generation. *Journal of Thermal Stresses*, 45(3):191–213, 2022

I Çömez, Y Alinia, and MA Güler. The partial slip contact problem for a monoclinic coated half plane. *Archives of Mechanics*, 74(1):13–37, 2022

Sami El-Borgi, Isa Çömez, and Mehmet Ali Güler. A receding contact problem between a graded piezoelectric layer and a piezoelectric substrate. *Archive of Applied Mechanics*, 91(12):4835–4854, 2021

E Nart, Y Alinia, and MA Güler. The effect of material anisotropy on the mechanics of a thin–film/substrate system under mechanical and thermal loads. *Mathematics and Mechanics of Solids*, page 10812865211031277, 2021

M Kathiresan, K Manisekar, Vasudevan Rajamohan, and Mehmet A Güler. Investigations on crush behavior and energy absorption characteristics of GFRP composite conical frusta with a cutout under axial compression loading. *Mechanics of Advanced Materials and Structures*, pages 1–18, 2021

Barış Mehmet Zeytinci, Mehmet Şahin, Mehmet Ali Güler, and Konstantinos Daniel Tsavdaridis. A practical design formulation for perforated beams with openings strengthened with ring type stiffeners subject to Vierendeel actions. *Journal of Building Engineering*, page 102915, 2021

Sabri Alper Keskin, Erdem Acar, MA Güler, and Murat Altin. Exploring various options for improving crashworthiness performance of rail vehicle crash absorbers with diaphragms. *Structural and Multidisciplinary Optimization*, pages 1–16, 2021

E Özaslan, A Yetgin, B Acar, and MA Güler. Damage Mode Identification of Open Hole Composite Laminates Based on Acoustic Emission and Digital Image Correlation Methods. *Composite Structures*, page 114299, 2021

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